

So you have some DATA
now what?

Starting to work with data

Describing your data: what
are you working with?

Some basic functions to
begin working with your
data

Lecture 02: Beginning to work with data

So you have some DATA -
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So you have some DATA - now what?

Learning objectives for today:

1. What is a data frame
2. Get the data into R
3. Figure out what's in the dataset
 - ▶ Identifying the unit of analysis
 - ▶ Differentiating between the types of variables
4. Manipulate the data frame using the R package dplyr's main functions:
 - ▶ `rename()`
 - ▶ `select()`
 - ▶ `arrange()`
 - ▶ `filter()`
 - ▶ `mutate()`
 - ▶ `group_by()`
 - ▶ `summarize()`

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Your lab this week will be about getting oriented in the R and R studio environment.

Here we will talk about some tasks you will do in R to start exploring a dataset

What do you have?

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1. What is a data frame
 - ▶ Identifying the unit of analysis
 - ▶ Differentiating between the types of variables
2. Get the data into R
3. Figure out what's in the dataset
4. Start to manipulate the dataset

What is a data frame?

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- ▶ A data frame is a data set.
- ▶ We read data into R from common sources like Excel spreadsheets (.xls or .xlsx), text files (.txt), comma separate value files (.csv), and other formats.
- ▶ The simplest format of data contains one row for each individual in the study.
- ▶ The first column of the data identifies the **individual** (perhaps by a name or an ID **variable**).
- ▶ Subsequent columns are **variables** that have been recorded or measured.

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The unit of analysis is the major entity you are working with:

- ▶ Bacteria
- ▶ Laboratory test results
- ▶ Individual People
- ▶ Groups of people (couples, households)
- ▶ Villages
- ▶ Countries

Type of Variable

- ▶ **Categorical** variable: A variable that has grouping levels. Mathematically you can calculate the proportion (%) of individuals in each level of the category.
 - ▶ **Nominal** variables: have no underlying order or rank. E.g., hospital ID, HIV status (yes/no variables), race
 - ▶ **Ordinal** variables: can be ordered or ranked. E.g., socio-economic status, BMI categories
- ▶ **Quantitative** variable: A continuous, numeric variable that you can perform mathematical operations on. Mathematically, we can you take the median or average of these variables
 - ▶ **Discrete** variables: can be counted. E.g., number of brain lesions, number of previous births
 - ▶ **Continuous** variables: can be measured precisely, with a ruler or scale. E.g, annual income, blood alcohol content, gestational age at birth

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Lake data from Baldi and Moore (B&M)

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- ▶ Exercise 1.25 from Edition 4 of B&M
- ▶ Data from a study of mercury concentration across 53 lakes
- ▶ I've placed these data in my working directory

readr is a package in R

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- ▶ you will learn more about packages in lab
- ▶ once the package has been “installed” you will need to call it into active use,
- ▶ To access readr’s functions we load the library like this:

```
library(readr)
```

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`read_csv()` to load the lake data in R

- ▶ `read_csv()` is a function from the `readr` library used to import csv files.
- ▶ code template: `your_data <- read_csv("pathway_to_data.csv")`
- ▶ The `<-` is called the **assignment operator**. It says to save the imported data into an object called `your_data`.

```
lake_data <- read_csv("mercury-lake.csv")
```

Four R functions to get to know a dataset

- ▶ `head(your_data)`: Shows the first six rows of the supplied dataset
- ▶ `dim(your_data)`: Provides the number of rows by the number of columns
- ▶ `names(your_data)`: Lists the variable names of the columns in the dataset
- ▶ `str(your_data)`: Summarizes the above information and more

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notice that if I put a # in front of a line of code it will not run

```
#head(lake_data)
```

```
#dim(lake_data)
```

```
#names(lake_data)
```

```
str(lake_data)
```

```
## spc_tbl_ [9 x 6] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ lakes      : chr [1:9] "Alligator" "Annie" "Apopka" "Blue Cypress" ...
## $ ph         : num [1:9] 6.1 5.1 9.1 6.9 4.6 7.3 5.5 7.3 8.2
## $ chlorophyll: num [1:9] 0.7 3.2 128.3 3.5 1.8 ...
## $ mercury    : num [1:9] 1.23 1.33 0.04 0.44 1.2 0.27 0.33 0.17 1.87
## $ number_fish: num [1:9] 5 7 6 12 12 14 5 8 3
```

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Some basic functions to begin working with your data

Function 1: rename()

What do you think rename does?

First print the names of the variables:

```
names(lake_data)
```

```
## [1] "lakes"          "ph"              "chlorophyll"    "mercury"        "number_fish"  
## [6] "age_data"
```

Run the rename() function and assign it to lake_data_tidy:

```
lake_data_tidy <- rename(lake_data, name_of_lake = lakes)
```

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Function 1: rename()

Then reprint the variable names:

```
names(lake_data_tidy)
```

```
## [1] "name_of_lake" "ph"  
## [6] "age_data"
```

```
"chlorophyll" "mercury"
```

```
"number_fi
```

Function 1: rename() multiple variables at once

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You can rename multiple variables at once:

```
lake_data_tidy <- rename(lake_data,  
                          name_of_lake = lakes,  
                          ph_level = ph)
```

Code template for rename() function

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```
new_dataset <- rename(old_dataset, new_name = old_name)
```

Another way to write the above code is to use the **pipe** operator: %>%

```
new_dataset <- old_dataset %>% rename(new_name = old_name)
```

The pipe will become very useful in a few slides...

Function 2: select()

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Based on the output below, what do you think select() does?

```
smaller_data <- select(lake_data, lakes, ph, chlorophyll)
names(smaller_data)
```

```
## [1] "lakes"          "ph"             "chlorophyll"
```

Function 2: `select()`

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- ▶ We use `select()` to select a subset of **variables**.
- ▶ This is very handy if we inherit a large dataset with several variables that we do not need.

Function 2: “negative select()”

We can also use “negative select()” to deselect variables. Suppose we wanted to keep all variables except for age_data:

```
smaller_data_2 <- select(lake_data, - age_data)  
names(smaller_data_2)
```

```
## [1] "lakes"          "ph"              "chlorophyll"    "mercury"        "number_fish"
```

We place a negative sign in front of age_data to remove it from the dataset.

Rewrite using the pipe operator

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```
smaller_data <- lake_data %>% select(lakes, ph, chlorophyll)  
smaller_data_2 <- lake_data %>% select(- age_data)
```

- ▶ Going forward, we will use the pipe operator to write code using any dplyr functions
- ▶ This is because we can use the pipe to stack many dplyr functions in a row

Function 3: arrange()

What does arrange do? First type `View(lake_data)` to look at the original data. Then run the code and examine its output below. What is different?:

```
if (interactive()) View(lake_data)
lake_data %>% arrange(ph)
```

```
## # A tibble: 9 x 6
##   lakes          ph chlorophyll mercury number_fish age_data
##   <chr>        <dbl>      <dbl>    <dbl>      <dbl> <chr>
## 1 Brick        4.6        1.8      1.2        12 year old
## 2 Annie        5.1        3.2     1.33         7 recent
## 3 Catalina     5.5       13.2     0.33         5 recent
## 4 Alligator    6.1         0.7     1.23         5 year old
## 5 Blue Cypress 6.9         3.5     0.44        12 recent
## 6 Bryant       7.3       44.1     0.27        14 year old
## 7 Four Mile    7.3         0.4     0.17         8 recent
## 8 Henry        8.2       12.2     1.87         3 year old
```


Function 3: arrange() in descending order

```
lake_data %>% arrange(- ph)
```

```
## # A tibble: 9 x 6
##   lakes          ph chlorophyll mercury number_fish age_data
##   <chr>        <dbl>      <dbl>    <dbl>      <dbl> <chr>
## 1 Apopka        9.1      128.     0.04         6 recent
## 2 Henry          8.2      12.2     1.87         3 year old
## 3 Bryant         7.3      44.1     0.27        14 year old
## 4 Four Mile      7.3       0.4     0.17         8 recent
## 5 Blue Cypress   6.9       3.5     0.44        12 recent
## 6 Alligator       6.1       0.7     1.23         5 year old
## 7 Catalina        5.5      13.2     0.33         5 recent
## 8 Annie           5.1       3.2     1.33         7 recent
## 9 Brick           4.6       1.8     1.2        12 year old
```

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Function 3: arrange() by two variables

```
lake_data %>% arrange(age_data, -ph)
```

```
## # A tibble: 9 x 6
##   lakes          ph chlorophyll mercury number_fish age_data
##   <chr>        <dbl>      <dbl>    <dbl>      <dbl> <chr>
## 1 Apopka        9.1      128.     0.04         6 recent
## 2 Four Mile      7.3        0.4     0.17         8 recent
## 3 Blue Cypress  6.9        3.5     0.44        12 recent
## 4 Catalina      5.5       13.2     0.33         5 recent
## 5 Annie         5.1        3.2     1.33         7 recent
## 6 Henry         8.2       12.2     1.87         3 year old
## 7 Bryant        7.3      44.1     0.27        14 year old
## 8 Alligator     6.1        0.7     1.23         5 year old
## 9 Brick         4.6        1.8     1.2        12 year old
```

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Function 4: mutate()

- ▶ `mutate()` is one of the most useful functions!
- ▶ It is used to add new variables to the dataset. Suppose that someone told you that the number of fish sampled was actually in hundreds, such that 5 is actually 500. You can use `mutate` to add a new variable to your dataset that is in the hundreds:

```
lake_data_new_fish <- lake_data %>%
  mutate(actual_fish_sampled = number_fish * 100)
```

```
lake_data_new_fish
```

```
## # A tibble: 9 x 7
```

```
##   lakes      ph chlorophyll mercury number_fish age_data actual_fish_
##   <chr>    <dbl>      <dbl>    <dbl>      <dbl> <chr>
## 1 Alligator 6.1        0.7      1.23        5 year old
## 2 Annie     5.1        3.2      1.33        7 recent
## 3 Apopka    9.1       128.      0.04        6 recent
```

Use %>% to append several lines of code together

- ▶ We have saved many of new datasets in our environment!
- ▶ If these datasets were larger, they would take up a lot of space.
- ▶ Rather than saving a new dataset each time, we can make successive changes to one dataset like this:

```
tidy_lake_data <- lake_data %>%  
  rename(name_of_lake = lakes) %>%  
  mutate(actual_fish_sampled = number_fish * 100) %>%  
  select(- age_data, - number_fish)  
tidy_lake_data
```

```
## # A tibble: 9 x 5  
##   name_of_lake    ph chlorophyll mercury actual_fish_sampled  
##   <chr>         <dbl>      <dbl>    <dbl>          <dbl>  
## 1 Alligator      6.1         0.7      1.23           500  
## 2 Annie          5.1         3.2      1.33           700  
## 3 Apopka         9.1        128.      0.04           600
```

Use %>% to “pipe” several lines of code together

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```
tidy_lake_data <- lake_data %>%  
  rename(lake_name = lakes) %>%  
  mutate(actual_fish_sampled = number_fish * 100) %>%  
  select(- age_data, - number_fish)
```

```
#tidy_lake_data
```

Function 5: `filter()`

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Filter is another very useful function! What might `filter()` do?

Function 5: `filter()`ing on numeric variables

We use `filter` to select which rows we want to keep in the dataset. Suppose you were only interested in lakes with `ph` levels of 7 or higher.

```
lake_data_filtered <- lake_data %>% filter(ph > 7)
lake_data_filtered
```

```
## # A tibble: 4 x 6
##   lakes      ph chlorophyll mercury number_fish age_data
##   <chr>    <dbl>      <dbl>    <dbl>      <dbl> <chr>
## 1 Apopka    9.1      128.     0.04         6 recent
## 2 Bryant    7.3      44.1     0.27        14 year old
## 3 Four Mile  7.3       0.4     0.17         8 recent
## 4 Henry     8.2      12.2     1.87         3 year old
```

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Function 5: filter()ing on character/string variables

Let's try a few more ways to filter() the data set since subsetting data is so important:

```
lake_data %>% filter(age_data == "recent")
```

```
## # A tibble: 5 x 6
##   lakes          ph chlorophyll mercury number_fish age_data
##   <chr>        <dbl>      <dbl>    <dbl>      <dbl> <chr>
## 1 Annie          5.1         3.2     1.33         7 recent
## 2 Apopka          9.1        128.     0.04         6 recent
## 3 Blue Cypress    6.9         3.5     0.44        12 recent
## 4 Catalina        5.5        13.2     0.33         5 recent
## 5 Four Mile       7.3         0.4     0.17         8 recent
```

► == is read as “is equal to”

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Function 5: filter()ing on character/string variables

```
lake_data %>% filter(age_data != "recent")
```

```
## # A tibble: 4 x 6
##   lakes      ph chlorophyll mercury number_fish age_data
##   <chr>    <dbl>      <dbl>    <dbl>      <dbl> <chr>
## 1 Alligator 6.1      0.7      1.23         5 year old
## 2 Brick     4.6      1.8      1.2        12 year old
## 3 Bryant    7.3     44.1     0.27        14 year old
## 4 Henry     8.2     12.2     1.87         3 year old
```

► != is read as “is not equal to”

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Function 5: filter()ing on character/string variables

```
lake_data %>% filter(lakes %in% c("Alligator", "Blue Cypress"))
```

```
## # A tibble: 2 x 6
##   lakes          ph chlorophyll mercury number_fish age_data
##   <chr>        <dbl>        <dbl>    <dbl>        <dbl> <chr>
## 1 Alligator    6.1          0.7      1.23          5 year old
## 2 Blue Cypress 6.9          3.5      0.44          12 recent
```

- ▶ %in% is the “in” operator. We are selecting rows where the variable lakes belongs to the specified list.
- ▶ The c() combines “Alligator” and “Blue Cypress” into a list

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Function 5: multiple filter()s at once

```
lake_data %>% filter(ph > 6, chlorophyll > 30)
```

```
## # A tibble: 2 x 6
```

```
##   lakes      ph chlorophyll mercury number_fish age_data
##   <chr>   <dbl>         <dbl>    <dbl>         <dbl> <chr>
## 1 Apopka  9.1         128.      0.04             6 recent
## 2 Bryant  7.3          44.1      0.27            14 year old
```

#this is the same as:

```
lake_data %>% filter(ph > 6 & chlorophyll > 30)
```

```
## # A tibble: 2 x 6
```

```
##   lakes      ph chlorophyll mercury number_fish age_data
##   <chr>   <dbl>         <dbl>    <dbl>         <dbl> <chr>
## 1 Apopka  9.1         128.      0.04             6 recent
## 2 Bryant  7.3          44.1      0.27            14 year old
```

Function 5: filter() using “or”

```
lake_data %>% filter(ph > 6 | chlorophyll > 30)
```

```
## # A tibble: 6 x 6
```

##	lakes	ph	chlorophyll	mercury	number_fish	age_data
##	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
## 1	Alligator	6.1	0.7	1.23	5	year old
## 2	Apopka	9.1	128.	0.04	6	recent
## 3	Blue Cypress	6.9	3.5	0.44	12	recent
## 4	Bryant	7.3	44.1	0.27	14	year old
## 5	Four Mile	7.3	0.4	0.17	8	recent
## 6	Henry	8.2	12.2	1.87	3	year old

- ▶ | is the OR operator. At least one of `ph > 6` or `chlorophyll > 30` needs to be true.

Functions 6 and 7: group_by() and summarize()

Let's execute the following code and see what it does.

```
lake_data %>%  
  group_by(age_data) %>%  
  summarize(mean_ph = mean(ph))
```

```
## # A tibble: 2 x 2  
##   age_data mean_ph  
##   <chr>      <dbl>  
## 1 recent      6.78  
## 2 year old    6.55
```

What happened?

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Functions 6 and 7: group_by() and summarize()

Another one:

```
lake_data %>%  
  group_by(age_data) %>%  
  summarize(mean_ph = mean(ph),  
             standard_deviation_ph = sd(ph))
```

```
## # A tibble: 2 x 3  
##   age_data mean_ph standard_deviation_ph  
##   <chr>      <dbl>                <dbl>  
## 1 recent      6.78                1.59  
## 2 year old    6.55                1.56
```

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Recap: What functions did we use?

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1. `library()` to load `readr` and `dplyr`.
2. `read_csv()` to read csv files from a directory.
3. `head()`, `str()`, `dim()`, and `names()` to look at our imported data.
4. `rename()` to rename variables in a data frame.
5. `select()` to select a subset of variables.
6. `arrange()` to sort a dataset according to one or more variables.
7. `mutate()` to create new variables.
8. `filter()` to select a subset of rows.
9. `group_by()` and `summarize()` to group the data by a categorical variable and calculate a statistic.
10. `mean()` and `sd()` to calculate the mean and standard deviation of variables.

Recap: What operators did we use?

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1. Assignment arrow: `<-`: This is our most important operator!
2. Greater than: `>` There are also:
 - ▶ Less than: `<`
 - ▶ Greater than or equal to: `>=`, and,
 - ▶ Less than or equal to: `<=`
3. Is equal to: `==`, and `!=` is not equal to
4. `%in%` to select from a list, where the list is created using `c()`, i.e., lakes
`%in% c("Alligator", "Annie")`

How to export from datahub and save onto your own computer

Some of you may want to edit this file in R markdown by adding notes, etc. In that case, you can make your edits on datahub and save your updated file on the cloud. You can additionally save your updated file locally on your computer. Here's how to do that:

1. In the File view window, click the checkbox beside the file you'd like to export
2. click More > Export.

This will download the file to your computer's downloads folder.

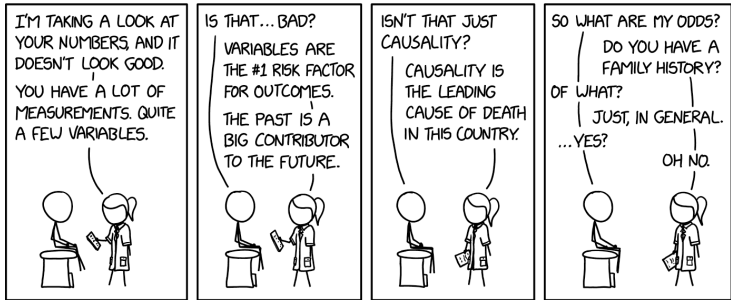
Parting humor

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courtesy of xkcd.com